

Supplementary File S1

Supplementary Materials for:

Data-Driven Forecasting of PISA Reading Literacy Performance of Türkiye: Artificial Intelligence Based Time Series Estimation

S1. Data Sources

The study uses publicly available data obtained from the OECD Programme for International Student Assessment (PISA) database. Reading literacy scores from selected countries and Türkiye were collected for the PISA cycles between 2003 and 2022.

In addition, four independent variables associated with student performance were included:

- Highest Parental Education (PARED)
- Cultural Possessions (CULTPOS)
- Highest Parental Occupational Status (HISEI)
- Sense of Belonging at School (BELONG)

S2. Variable Definitions

Variable	Description
Reading Literacy	Average PISA reading literacy score
PARED	Highest parental education in years
CULTPOS	Cultural possessions index
HISEI	Highest parental occupational status index
BELONG	Sense of belonging at school index

S3. Missing Data Treatment

Reading Literacy Scores

Two countries contained missing observations:

Country	Missing Cycle	Method
Malaysia	2006	Linear interpolation
Peru	2006	Linear interpolation

Independent Variables

Variable	Missing Cycle(s)	Method
CULTPOS	2022	Linear extrapolation

Variable	Missing Cycle(s)	Method
BELONG	2003, 2006, 2009	Backward extrapolation

The missing CULTPOS value for 2022 was estimated using the decreasing trend observed between 2015 and 2018.

The missing BELONG values were estimated using backward extrapolation based on the trend observed between 2012 and 2015.

S4. Data Standardization

All variables were standardized using Z-score normalization before model training.

$$z = (x - \mu) / \sigma$$

where x represents the original value, μ represents the mean, and σ represents the standard deviation.

S5. Sliding Window Transformation

A sliding window approach was used to transform the time series into supervised learning sequences.

Input sequence:

$$X_t = [x_{t-n+1}, x_{t-n+2}, \dots, x_t]$$

Output sequence:

$$x_{t+1}$$

The window was recursively shifted across the time series to generate training samples.

S6. Forecasting Strategy

Recursive multi-step forecasting was employed.

Forecasting steps:

1. Forecast 2025 using data up to 2022.
2. Forecast 2028 using observed values and the 2025 forecast.
3. Forecast 2031 using observed values and forecasts up to 2028.

S7. Model Configuration

The same hyperparameter configuration was used across deep learning models.

Hyperparameter	Value
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Hyperparameter	Value
Units	50
Learning Rate	0.001
Epochs	50
Batch Size	16
Dropout	0.0
Optimizer	Adam

S8. Evaluation Metrics

Model performance was evaluated using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Dynamic Time Warping (DTW)

S9. Sequential Variable Inclusion Strategy

Independent variables were sequentially incorporated into the forecasting models according to their estimated contribution:

Model 0: Reading Literacy

Model 1: Reading Literacy + PARED

Model 2: Reading Literacy + PARED + CULTPOS

Model 3: Reading Literacy + PARED + CULTPOS + HISEI

Model 4: Reading Literacy + PARED + CULTPOS + HISEI + BELONG

S10. Shared Materials

The supplementary package contains:

- Raw datasets
- Processed datasets
- Final model dataset
- Forecast outputs
- Python source code
- README documentation
- Software requirements

These materials are provided to facilitate transparency, reproducibility, and future research.